CS-471: Machine Learning

Week 7

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- CART Algorithm
- Quiz

Recap of Previous Lecture

- What stopping criteria are used in decision trees?
- What to do if after using all attributes we still have examples that do not belong to the same class?
- What to do if there are no examples in a particular subset?
- How to split if we have continuous valued attributes?
- How do we make decision boundaries with continuous valued attributes?
- What is ID3 algorithm used for?
- What splitting criteria is used in ID3?

Recap of Previous Lecture

- What is the stopping criteria in ID3?
- What are the limitations of ID3 algorithm?
- What is C4.5 algorithm used for?
- What is its advantage over ID3?
- What splitting criteria does C4.5 use?
- How is Gain Ratio determined?
- What is CART algorithm used for? Kind of partitioning?
- What splitting criteria does CART use? Stopping criteria?

How C4.5 handles missing values

- During Split Selection:

$$Gain(A) = \frac{N_{known}}{N} \times Gain\ on\ known\ subset$$

- During Tree Construction:
  - Samples with missing attribute values are distributed fractionally

- During Prediction:
  - Sample with missing attribute goes through multiple branches and final prediction is a weighted sum of branch probabilities of different classes

Example

Student	Study Hours	Attendance (%)	Pass (1) or Fail (0)
A	1	50	0
B	5	80	1
C	6	85	1
D	2	40	0
E	3	60	0
F	8	90	1

63P, 37F = 100 values

$S \leq 4$

Yes

60 values = 27P, 33F

$A \leq 60$

Yes

6P, 24F

Fail

NO

21P, 9F

Pass.

NO

36P, 4F = 40 values.

Pass.

CART Algorithm for Regression

- Start with the entire dataset.
- Choose the best feature to split using MSE.
- Recursively split dataset into smaller binary subsets.
- Stop when a stopping criterion is met.
- Assign predictions as the mean value in each leaf node.

CART for Regression Example

Student	Study Hours	Attendance (%)	Marks
A	1	50	35
B	5	80	75
C	6	85	82
D	2	40	40
E	3	60	55
F	8	90	92

Questions?